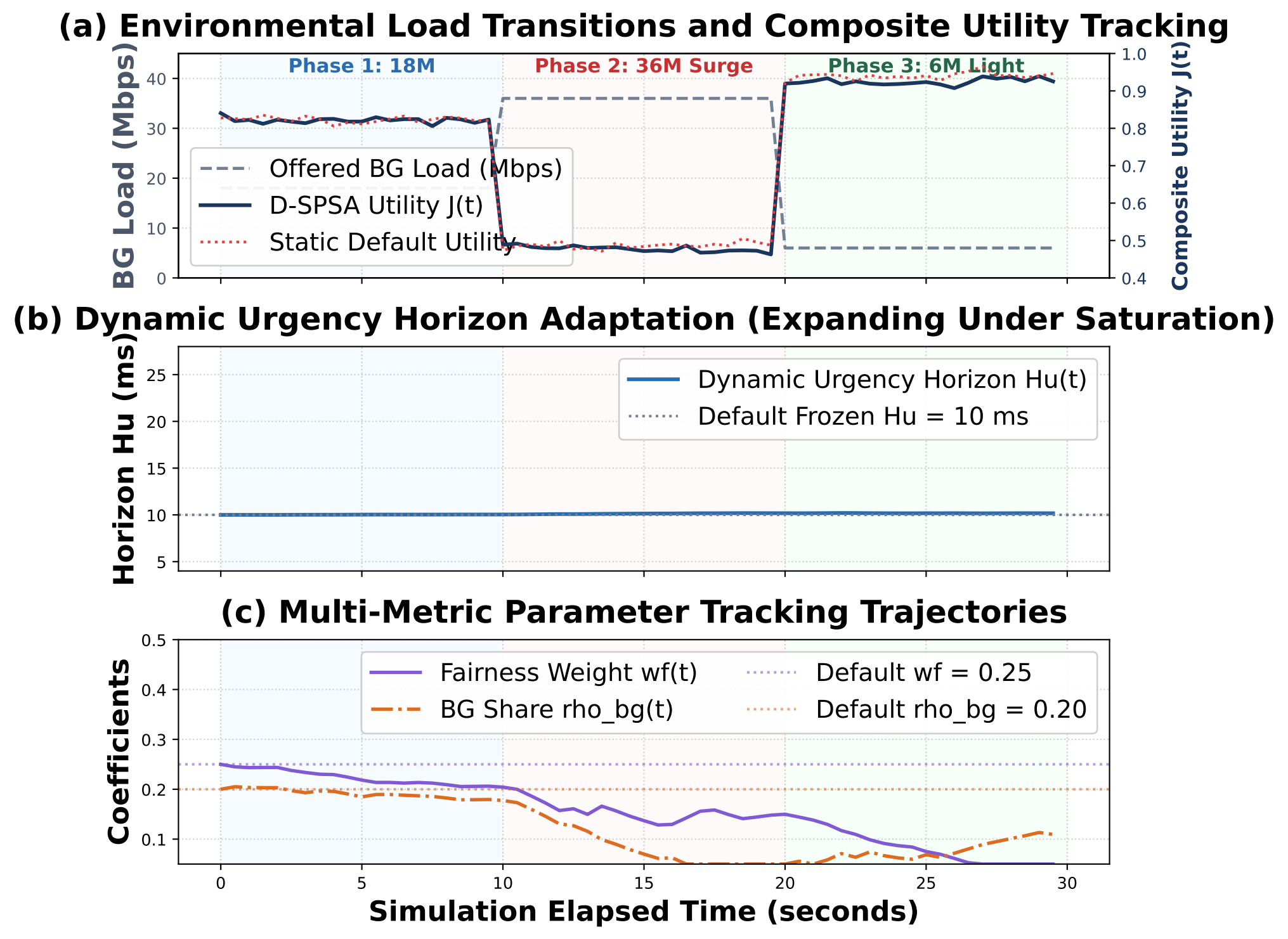
(a) Environmental Load Transitions and Composite Utility Tracking
Phase 1: 18M
Phase 2: 36M Surge
Phase 3: 6M Light
BG Load (Mbps)
Composite Utility J(t)
Offered BG Load (Mbps)
D-SPSA Utility J(t)
Static Default Utility
(b) Dynamic Urgency Horizon Adaptation (Expanding Under Saturation)
Horizon Hu (ms)
Dynamic Urgency Horizon Hu(t)
Default Frozen Hu = 10 ms
(c) Multi-Metric Parameter Tracking Trajectories
Coefficients
Fairness Weight wf(t)
BG Share rho_bg(t)
Default wf = 0.25
Default rho_bg = 0.20
Simulation Elapsed Time (seconds)